

ITZEHOE, Germany

WMO: 101420

Lat: 53.9897N

Lon: 9.5697E

Elev: 84

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.7	18.8	10.8	9.6	17.0	14.8	11.7	20.4	28.5	46.0	25.5	44.5	4.8	120	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	83.2	65.7	79.0	65.0	75.5	63.9	68.5	78.3	66.8	75.6	65.0	73.0	7.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.0	72.3	63.5	87.8	70.4	61.7	82.5	68.6	32.8	78.5	31.4	75.7	30.0	73.8	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.0	18.1	15.9	DB	7.3	90.5	6.3	3.0	2.7	92.6	-1.0	94.4	-4.5	96.1	-9.2	98.3
			WB	7.4	72.7	6.5	1.5	2.8	73.8	-1.0	74.7	-4.7	75.5	-9.4	76.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	34.4	35.3	38.7	47.1	53.5	58.2	63.1	63.1	57.2	49.4	41.5	35.1	(6)
	DBStd	12.38	8.00	7.70	7.17	6.87	5.96	5.72	5.51	5.22	5.27	6.24	6.65	7.90	(7)
	HDD50	2232	482	413	351	137	31	2	0	0	4	87	262	463	(8)
	HDD65	6274	947	832	814	538	361	218	105	102	241	482	706	928	(9)
	CDD50	1545	0	1	3	48	138	249	406	405	220	69	6	0	(10)
	CDD65	112	0	0	0	0	3	15	46	42	6	0	0	0	(11)
	CDH74	1080	0	0	0	9	49	170	450	355	46	1	0	0	(12)
	CDH80	266	0	0	0	0	7	41	120	95	3	0	0	0	(13)
(14) Wind	WSAvg	7.1	8.2	8.9	8.6	7.2	6.9	6.5	6.2	5.9	6.0	6.5	6.8	7.2	(14)
(15) Precipitation	PrecAvg	33.9	2.8	2.4	2.2	1.6	2.4	3.2	3.6	3.5	3.1	2.8	3.0	3.3	(15)
	PrecMax	43.4	6.3	5.8	4.8	3.7	5.6	5.9	9.1	8.9	7.5	8.6	6.1	7.6	(16)
	PrecMin	23.9	0.1	0.4	0.3	0.1	0.5	0.2	1.3	1.1	0.7	1.2	0.4	1.0	(17)
	PrecStd	4.9	1.5	1.2	1.1	0.8	1.1	1.4	2.0	1.7	1.7	1.5	1.6	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	53.5	61.3	74.6	79.5	85.1	89.0	87.2	79.1	67.5	58.1	53.1	(19)
		MCWB	49.5	48.2	51.6	58.4	62.5	65.8	68.6	68.1	65.2	60.5	54.4	50.5	(20)
	2%	DB	49.7	50.4	55.5	68.8	74.0	79.1	83.2	82.8	74.1	63.2	54.9	49.9	(21)
		MCWB	47.0	46.8	49.2	55.9	59.7	64.5	66.4	65.7	63.7	58.2	52.1	47.9	(22)
	5%	DB	47.3	47.8	52.0	64.1	69.9	73.9	79.1	77.7	69.5	60.5	52.7	47.5	(23)
		MCWB	45.3	45.2	47.6	53.2	57.7	62.4	65.5	65.1	61.2	56.3	50.3	45.6	(24)
	10%	DB	45.0	45.4	48.9	59.2	65.8	69.5	75.1	73.5	66.0	58.4	50.6	45.5	(25)
		MCWB	43.3	43.2	45.1	51.0	56.1	60.0	64.5	63.9	59.1	55.0	48.5	43.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	50.5	53.3	59.5	65.4	69.5	71.0	71.4	67.2	61.4	55.3	51.0	(27)
		MCDB	52.3	52.4	57.3	71.2	75.2	80.4	80.8	82.9	74.6	66.0	57.2	52.7	(28)
	2%	WB	47.3	47.3	50.3	56.5	62.1	66.4	68.8	68.1	64.5	58.8	52.8	47.9	(29)
		MCDB	48.8	49.7	54.6	66.5	70.4	75.7	79.1	76.8	72.7	62.2	54.8	49.5	(30)
	5%	WB	45.5	45.2	47.8	54.2	59.0	63.4	67.0	66.3	62.0	57.1	50.7	45.7	(31)
		MCDB	47.0	47.5	51.6	62.7	66.7	70.7	76.2	74.8	67.9	59.9	52.4	47.1	(32)
	10%	WB	43.3	43.2	45.5	51.8	56.8	61.1	65.0	64.6	60.2	55.4	48.7	44.0	(33)
		MCDB	45.1	45.2	48.5	58.1	64.3	68.0	73.0	72.0	65.2	58.0	50.4	45.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.9	12.3	17.2	18.2	16.7	17.1	16.5	15.1	12.6	9.0	8.1	(35)
		MCDBR	8.4	9.3	17.4	24.9	25.7	25.0	25.2	23.5	20.5	14.0	8.9	8.3	(36)
	5% WB	MCWBR	7.6	7.4	12.3	13.9	14.5	13.0	11.6	11.1	11.8	9.6	7.4	7.6	(37)
		MCWBR	8.2	8.6	15.5	21.8	21.5	20.8	21.8	20.0	18.0	12.8	8.5	8.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.320	0.338	0.367	0.412	0.404	0.402	0.420	0.414	0.388	0.356	0.334	0.311	(40)
		taud	2.318	2.329	2.277	2.195	2.260	2.291	2.258	2.286	2.342	2.409	2.363	2.312	(41)
	Ebn at Noon		200	237	255	258	267	269	261	255	246	228	193	178	(42)
		Edh at Noon	19	25	33	40	40	39	40	36	31	23	18	16	(43)
(44) All-Sky Solar Radiation	RadAvg		172	380	772	1284	1617	1685	1597	1310	942	515	215	128	(44)
	RadStd		20	54	74	150	170	136	201	98	107	58	24	9	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-256	N/A	N/A

Nomenclature: See separate page